GEOPHYSICAL RESEARCH LETTERS, VOL. 13, NO. 13, PAGES 1442-1445, DECEMBER 1986

NORMAL AMPLITUDE BRUNHES PALEOSECULAR VARIATION AT LOW-LATITUDES: A PALEOMAGNETIC RECORD FROM THE TRANS-MEXICAN VOLCANIC BELT

Emilio Herrero-Bervera[1], Jaime Urrutia Fucugauchi[2], Ana Lillian Martin Del Pozzo[2], Harald Böhnel[2], and José Guerrero[3]

[1] Hawaii Institute of Geophysics, University of Hawaii, Honolulu, Hawaii 96822

[2] Laboratorio de Paleomagnetismo, Instituto de Geofísica, U.N.A.M., México, D.F. 04510, MEXICO

[3] Instituto de Geología, U.N.A.M., México D.F. 04510, MEXICO

Abstract. Ancient secular variation in Central Mexico was determined from paleomagnetic measurements on 45 independent lava flows ranging in age (^{14}C and K-Ar dates) from 2,500 to 580,000 years B.P. All the analyzed flows are characterized by a normal polarity placing them within the Brunhes Chron. The paleosecular variation analysis yielded an angular standard deviation from the field of an axial dipole of 15.3° with 95% confidence limits of 13.4° and 17.9°, and that of the corresponding Virtual Geomagnetic Poles (VGPs) of 13.9° with confidence limits of 12.1° and 16.1°. These values are in agreement with those predicted by most statistical models of secular variation, but divergent from those obtained from low latitudes (19°N) such as Hawaii and Pagan Is.

Introduction

During historic time, both the amount of secular variation of the Earth's magnetic field (SV) and the magnitude of the nondipole field, whose changes cause most of the historically observed secular variation, have been rather irregularly distributed over the earth's surface [Cox and Doell, 1964; Doell and Cox, 1972]. These features are at the present time of larger magnitude in the Southern Hemisphere than in the Northern Hemisphere and are almost completely lacking in the central Pacific region [Doell and Cox, 1962; McWilliams et al., 1982]. As a result of evidence from Hawaii that suggested a sustained interval of low geomagnetic secular variation, analogous studies were conducted elsewhere in the Pacific on the Galapagos [Doell and Cox, 1962], on Easter Island [Issacson and Heinrichs, 1976], and in the Society Islands [Duncan, 1975]. These studies showed secular variation consistent with a "standard" model of global variation based upon modern dipole and nondipole fields [Cox, 1970; McWilliams et al., 1982] and hence the region of anomalously low secular variation has been constrained to smaller regions of the Pacific. Another area characterized by a low geomagnetic secular variation is the Pagan Island in the Marianas (18.1°N, 145.7° E) in the Western Pacific. Three possible interpretations of this paleomagnetic study of 24 lava flows are: 1) the sites studied represent an inadequate coverage in time; 2) the Marianas should be included in the Pacific dipole window; 3) there is a belt of low average angular dispersion encircling the Earth between (roughly) 0 and 40° North [U.S.-Japan Paleomagnetic Cooperation Program in Micronesia, 1975].

The past distribution of such features and their permanence are of interest in determining, ultimately, boundary conditions for theories of the dynamo processes in the Earth's core that give rise to the geomagnetic field. Detailed sequential changes in geomagnetic field directions can rarely be determined paleomagnetically [Doell, 1970]; for this reason the amount of angular dispersion measured about some assumed (or mean) field distribution becomes an important quantity for a given locality.

Until now no data set has been sufficiently complete to be used to constrain secular variation in Mexico, and particularly in the Trans-Mexican Volcanic Belt (TVB). It is possible that the anomalously low secular variation observed in the Pacific could be expanded into the TVB. Doell and Cox's [1962] study shows that the intensity F_N of the nondipole 1965 International Geomagnetic Reference Field (IGRF) is less than 4000 nT in the central part of Mexico. On the other hand, if one believes one of the conclusions of the Pagan [U.S.-Japan Paleomagnetic Cooperation Program in Micronesia, 1975] study that there is a belt of low average angular dispersion encircling the earth (roughly) 0 and 40° N., then hypothetically the Chichinautzin Formation (located at 19° N) should be included as part of such a belt. Thus, pursuing a paleomagnetic study of the paleosecular variation from low latitudes is a worthy goal.

^{14}C and K-Ar dates

The study of the paleosecular variation of lava flows from the Sierra del Chichinautzin is one of several studies needed to understand the long-term nature of paleosecular variation at low latitudes. The time span selected for these studies is the Brunhes normal polarity Chron, which extends approximately from 0.73 my to the present. The Trans-Mexican Volcanic Belt (TVB) is a 20-70 km-broad structure which extends roughly East-West from Veracruz on the coast of Gulf of Mexico to near Puerto Vallarta in the West [Mooser et al., 1974]. The Chichinautzin Formation is part of the TVB and is located just South of Mexico City enclosing the southern margin of the Valley of Mexico. It is known from

Copyright 1986 by the American Geophysical Union.

Paper number 6L7053.
0094-8276/86/006L-7053$03.00

TABLE 1. Flow-Average Data for the Chichinautzin Formation Volcanics

Unit No.	No. of Samples	a.f. Cleaning (Peak field) mT	RM Declination °E	RM Inclination deg. Down	Length of Resultant R	Precision Parameter k	α_{95} Confidence	Angular s.d. deg. δ	VGP Lat., °N	VGP Long., °E	Age Unit	References
CHI-13	8	NRM	355.1	38.9	7.938	113.5	5.2	7.1	84.7	158.5	Recent	1
CHI-14	8	NRM	356.8	52.3	7.954	151.2	4.5	6.2	76.1	110.6	Recent	1
CHI-15	7	NRM	356.1	33.6	6.903	62.1	7.7	9.5	86.2	202.3	Recent	1
CHI-6	17	10	358.0	34.3	16.947	300.9	2.1	4.5	88.0	202.7	Hv	1
CHI-7	11	25	354.6	33.7	10.884	86.0	5.0	8.3	84.8	197.6	Hv	1
CHI-8	9	25	7.2	33.4	8.843	51.0	7.3	10.7	83.1	2.6	Hv	1
CHI-9	8	55	4.5	27.1	7.939	115.0	5.2	7.1	83.5	321.2	Hv	1
CHI-10	9	15	6.0	34.9	8.932	118.1	4.8	7.0	84.3	10.4	Hv	1
MO-10	6	60	356.9	34.9	5.945	92.0	7.0		87.1	172.0	Hv	2
MO-34	7	15	352.8	41.8	6.988	489.0	2.7		81.7	208.3	Hv	2
JU	7	50	323.2	22.4	6.979	289.2	3.5	6.3	53.8	164.5	Hv	This Study
XA	5	75	343.8	22.1	4.984	241.8	4.9	3.5	72.6	146.9	Hv	This Study
P-1	6	25	5.0	23.7	5.931	72.4	7.9	4.2	81.6	296.3	Hv	This Study
P-2	8	25	3.0	25.7	7.887	62.2	7.1	7.4	83.6	287.5	Hv	This Study
P-3	8	30	355.9	29.9	7.896	67.3	6.8	9.2	84.9	132.5	Hv	This Study
P-4	8	25	357.6	28.4	7.830	41.2	8.7	11.8	85.3	111.0	Hv	This Study
MO-5	8	60	346.5	38.3	7.938	114.0	5.2		77.1	183.8	Hv	2
MO-8	6	60	352.0	23.9	5.977	219.0	4.5		79.8	131.0	Plv4	2
MO-36	7	7.7	346.6	36.1	6.941	101.0	6.0		77.4	176.9	Plv4	2
TEU-1	7	NRM	338.7	17.1	6.969	199.8	4.2	9.3	66.9	147.2	Plv4	This Study
TEU-2	5	50	358.2	22.4	4.958	95.4	7.8	7.4	82.3	93.8	Plv4	This Study
CHI-11	12	10	16.1	36.1	11.952	229.7	2.9	5.1	74.8	14.6	Plv3	1
CHI-1	8	55	357.6	26.7	7.904	72.7	6.5	8.9	84.5	253.9	Plv1	1
CHI-2	14	20	6.1	12.9	13.937	205.4	2.8	5.5	76.2	304.9	Plv1	1
CHI-3	12	15	330.3	46.6	11.981	576.7	1.8	3.2	61.5	165.4	Plv1	1
CHI-4	9	20	342.7	11.6	8.950	160.2	4.1	6.0	68.6	224.9	Plv1	1
CHI-5	11	20	345.4	18.6	10.915	118.0	4.2	7.1	72.9	221.4	Plv1	1
MO-28	7	30	358.5	33.4	6.976	253.0	3.8		88.3	137.3	Plv1	2
OZ	4	50	352.8	23.1	3.992	411.8	4.5	5.7	80.1	125.9	Plv1	This Study
ACO	7	50	357.6	32.6	6.926	81.9	6.7	8.1	87.3	138.5	Plv1	This Study
IZT-82	8	60	356.1	39.8	7.960	178.0	4.2	5.7	85.0	215.4	80000 yr	3
IZT-29	7	30	7.4	38.1	6.920	72.6	7.1	8.8	82.7	332.3	80000 yr	3
IZT-30	7	120	356.6	31.9	6.920	79.4	6.8	8.4	86.3	141.6	80000 yr	3
IZT-31	7	60	3.8	35.9	6.910	64.3	7.6	9.4	86.4	339.4	80000 yr	3
IZT-10	6	60	7.4	28.5	5.970	199.3	4.8	5.2	81.9	19.7	270000 yr	3
IZT-13	7	120	355.9	31.5	6.990	569.1	2.5	3.2	85.6	143.2	270000 yr	3
IZT-78	7	30	1.3	28.3	6.980	243.2	3.9	4.8	85.7	64.4	340000 yr	3
IZT-11	7	60	3.6	23.5	6.960	168.4	4.7	5.8	82.3	54.3	410000 yr	3
IZT-18	7	60	17.3	52.9	6.910	65.6	7.5	9.3	69.0	305.1	410000 yr	3
IZT-20	7	60	22.7	60.8	6.960	158.4	4.8	6.0	60.3	296.9	410000 yr	3
IZT-21	7	60	352.4	32.1	6.950	131.7	5.3	6.5	82.6	158.9	410000 yr	3
IZT-24	7	30	355.3	21.3	6.930	82.4	6.7	8.3	80.7	111.2	410000 yr	3
IZT-25	6	120	27.0	61.5	5.980	327.6	3.7	4.1	57.3	299.6	410000 yr	3
IZT-133	6	60	10.2	25.5	5.980	207.5	4.7	5.1	78.6	20.4	580000 yr	3
IZT-27	6	60	354.0	30.0	5.940	79.6	7.6	8.3	83.5	144.0	580000 yr	3
MEAN	45		357.0	31.4	43.600	31.4	3.9	14.3	86.4	133.0	dp=2.4	dm=4.3

N=Number of Samples; a.f Cleaning= demagnetizing field in milliTeslas; RM Declination= Remanent Magnetization in degrees; RM Inclination= Remanent Magnetization Inclination in degrees; R=Length of Resultant Vector; k=Fisher's precision parameter; α_{95}= semiangle of cone of 95% confidence for mean directions; δ=Angular Standard deviation in degrees; VGP= Virtual Geomagnetic Pole Latitude and Longitude in degrees, respectively; and (dp,dm), oval of 95% confidence about the pole position. References: 1, Herrero-Bervera and Pal, [1978]; 2, Mooser et al., [1974]; 3, Steel, [1985]. Age Units: Recent=2500 yr; Hv=8500 yr; Plv4=19500 yr; Plv3=21860 yr; Plv1=38590 yr. Ages based on Martin Del Pozzo, [1982].

previous paleomagnetic studies [Mooser et al., 1974; Herrero-Bervera and Pal, 1977] that the age of the Chichinautzin Formation is not older than 700,000 years, and that the youngest of the lava eruptions have been in historic times. Charcoal samples were collected from the pottery level below the "Pedregal de San Angel" lavas near the pyramid of Cuicuilco. According to the radiometric determination reported [Libby, 1955], the samples were associated with pottery and figurines of Late Archaic (Qicoman phase type), yielding an age of 2422 +/- 250 years. These lava flows extend approximately 13 to 14 km towards the north from their source at the Xitle Volcano and represent the last volcanic activity in historic times.

The East part of the Chichinautzin Formation, has recently been studied in detail [Martin Del Pozzo, 1980; Martin Del Pozzo, 1982] and one hundred and forty six volcanoes, together with their deposits covering 952 square kilometers, were mapped. Cone density is 0.15 per square km with heights ranging from 15 to 315 m and crater diameters from 50 to 750 m. Ratios of cone height/diameter decreased from 0.20 to 0.12 with age. Basal diameters varied from 0.1 km to 2 km. Lengths of flows range from 1.0 to 21.5 km with heights of 0.5 to 300 m and aspect ratios of 21.4 to 350 [Martin Del Pozzo, 1982]. Thus the volcanoes from the Eastern part of the Chichinautzin Formation are comparable with the monogenetic volcanoes of the Western part. The Western part is characterized by ages ranging from 9.4 to 34.0 kyr [Bloomfield, 1975] determined by ^{14}C. In summary, the entire Chichinautzin Formation ranges from 2,400 to 700,000 years of age according to the ^{14}C (2,400 to 38,000 yrs.), K-Ar determinations and geomorphological features of the monogenetic volcanoes. Moreover, recent K-Ar studies from the Iztaccihuatl volcano, located on the easternmost part of the Chichinautzin Formation, give ages ranging from 580,000 +/- 110,000 to 76,000 +/- 20,000 years B.P. [Steele, 1985;

19448007, 1986, 13, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL013i013p01442 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

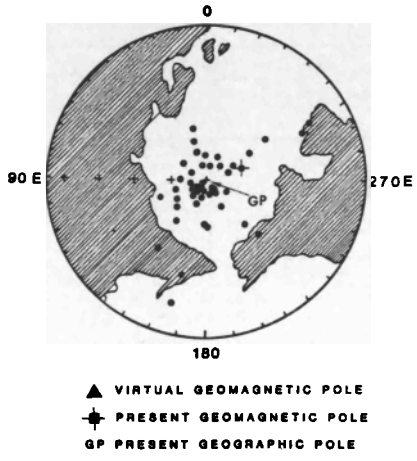


Fig. 1. Virtual Geomagnetic Pole positions of lavas in the Chichinautzin Formation. Cross is the present geomagnetic pole determined from the present field direction in the central part of Mexico.

Nixon et al., 1986]. Thus, all these dates place a lower limit of approximately 700,000 years on the entire Chichinautzin Formation.

In the Chichinautzin Sierra, volcanic activity has been sufficiently prolonged in the Brunhes interval to ascertain the overall pattern of recent secular variation without being biased by a single short burst of activity in the region. The present known age range is several times longer than the maximum periods (10,000 years) suspected to be involved in secular variation [Cox and Doell, 1964; McWilliams et al., 1982; Lund, 1985]. In the Chichinautzin Formation a time span from 2,500 to 580,000 years B.P. is available based on the radiocarbon and K-Ar dates. However, the data at hand clearly restrict the time period to the Brunhes normal polarity Chron.

Paleomagnetic Results

Our approach to the paleosecular variation problem in the central part of the TVB is based on the radiocarbon and K-Ar dates published for the Chichinautzin Formation, as well as our own paleomagnetic determinations for the area, supplemented by paleomagnetic data published by several workers in the area under study [Mooser et al., 1974; Herrero-Bervera and Pal, 1977; Steele, 1985].

Table 1 shows the results of the ^{14}C and K-Ar dates correlated with respect to the paleomagnetic results of the selected data that provided a good base to determine the paleosecular variation dispersion. Thus, based on the geology of the area and the Fisher statistics, we have made the attempt to group 45 lava flows [Baag and Helsley, 1974] out of 69 original lava flows; with alphas 95s of less than 9° [Urrutia-Fucugauchi, 1986] in order to conduct the dispersion statistics analysis of the flow-average data. Figure 1 shows the Virtual Geomagnetic Poles of the 45 selected lava flows.

Discussion

The quantity of main interest in this study is the angular dispersion which can be given by either the total standard deviation, S_T, or Fisher's precision parameter, K_T. The total disperson consists of two components: the between-site dispersion (K_B or S_B) and the within-site dispersion of the lava flows (K_W or S_W). The method used to calculate the between-site and within-site dispersions was the same as the one followed by Doell, [1970]. It is preferable to use S_B [Cox, 1969a; Doell, 1972b], which should not be affected by within-site scatter, rather than S_T. This method has been followed in this study. Table 2 gives the statistics obtained in this study. The angular standard deviation from the field of an axial dipole is 15.3° with Alpha 95 confidence limits [Cox, 1969a] of 13.4° and 17.9°, and that of the corresponding VGPs is 13.9°, with confidence limits of 12.1° and 16.1°. The global pattern of angular dispersion estimates the degree of dipole wobble and the latitudinal variation in relative strengths of the dipole and nondipole fields [McElhinny and Merrill, 1975]. It is important to mention that the other statistical parameter ΔI, which is the difference in site mean inclination (I) from the axial-dipole field expectation for this study is -3.46°. The global pattern of the ΔI anomaly estimates how well the axial-dipole hypothesis fits the actual geomagnetic field behavior [Lund, 1985].

Conclusions

Figure 2 gives a summary of several secular variation studies completed on 45 separate Brunhes-age lava flows. It is interesting to note that the angular dispersion $S_F=13.8^{\circ}$ with respect to the mean VGP is a close value to the value proposed for the normal polarity data for the last 5 Ma with, for example, model "F" ($S_F=13.7^{\circ}$) [McFadden and McElhinny, 1984] for 19 degrees North. The values obtained from the 45 selected lava flows from the Chichinautzin Formation are significantly different from the values reported for the angular dispersion found in Hawaii [McWilliams et al., 1982] and Pagan [U.S.-Japan Paleomagnetic Cooperation Program in Micronesia, 1975]. Both are in the "dipole window", but the portion of the TVB reported in this study should not be included in the "dipole window" even though the Chichinautzin Formation

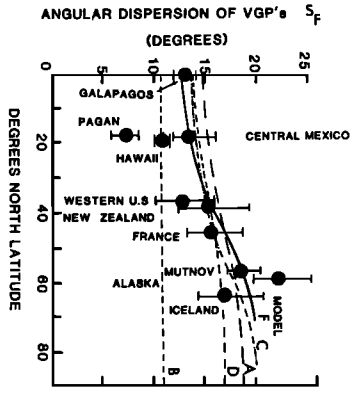


Fig. 2. Angular standard deviation for secular variation from various paleomagnetic data sets. The solid line shows the fit of model "F" [McFadden and McElhinny, 1984] to the 5 Myr average results.

19448007, 1986, 13, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL013i013p01442 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Dispersion Statistics of Flow Average Data

Valley of México volcanics ($\bar{N}$=7.7)	Angular Standard Deviations					Precision Parameters			
	S_T	S_W	S_B	S_F	α_{95}	k_T	k_W	k_B	k_F
Referred to:									
Mean Direction	14.6	2.1	14.6	14.6	3.8	31.1	112.5	37.2	37.6
Axial Dipole Direction	15.3	2.1	15.3	15.3	3.8	31.1	112.5	37.2	37.6
	S_T	S_W	S_B	S_F	α_{95}	K_T	K_W	K_B	K_F
Mean VGP	13.7	2.1	13.6	13.6	3.5	35.7	112.5	37.2	37.6
Geographic Pole	13.9	2.1	13.9	13.8	3.5	35.7	112.5	37.2	37.6

S=angular standard deviations for RM directions and VGP's, respectively; k,K=precision parameters for RM directions and VGP's, respectively, α_{95}=semiangle at cone of 95% confidence for mean directions and VGP's respectively, Subscripts: T=total dispersion, W=within flow dispersion, B=between flow dispersion (experimental errors only, removed), F=dispersions of VGP's and directions (experimental errors and effects of local anomalies removed); N=mean number of samples per lava used in the analysis.

is located at approximately the same latitude as Hawaii and Pagan Is.

Acknowledgements. The authors express their thanks to Dr M.W. McElhinny and one anonymous editor for reviewing this paper. This is a Hawaii Institute of Geophysics contribution no. 1790.

References

Baag, C., and C.E. Helsley, Shape analysis of paleosecular variation data, J. Geophys. Res. 79, 4923-4932, 1974.

Bloomfield, K., A Late-Quaternary monogenetic volcano field in central Mexico, Geol. Rundsch., 64, 476-497, 1975.

Cox, A., and R. R. Doell, Long period variation of the Geomagnetic field, Bull. Seismol. Soc. Amer., 54, 2243-2270, 1964.

Cox, A., Confidence limits for the precision parameter K, Geophys. J. R. Astr. Soc., 18, 545-549, 1969a.

Cox, A., Latitude dependence of the angular dispersion of the geomagnetic field, Geophys. J. Roy. Astron. Soc., 20, 253-269, 1970.

Doell, R. R., Paleomagnetic secular variation study of lavas from the Massif Central, France, Earth Planet. Sci. Lett., 8, 352-362, 1970.

Doell, R. R., Paleomagnetism of lava flows from Kauai, Hawaii, J. Geophys. Res., 77, 862-867, 1972b.

Doell, R. R. and A. Cox, The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle, in The Nature of Solid Earth, 245-284, McGraw-Hill, New York, 1972.

Duncan, R.A., Paleosecular variation at the Society Islands, French Polynesia, Geophys. J. Roy. Astron. Soc., 41, 245-254, 1975.

Herrero-Bervera, E. and Pal S., Paleomagnetic study of Sierra del Chichinautzin, Mexico, Geofísica Internacional, 17 (2), 167-180, 1977.

Issacson, L. B., and D. F. Heinrichs, Paleomagnetism and secular variation of Easter Island basalts, J. Geophys. Res., 81, 1476-1482, 1976.

Libby, F. W., Radiocarbon Dating, University of Chicago, 1955.

Lund, S. P., A comparison of the statistical secular variation recorded in some Late Quaternary lava flows and sediments, and its implications, Geophys. Res. Lett., 12, 251-254, 1985.

Martin Del Pozzo, A.L., Vulcanologia de la Sierra del Chichinautzin, Tesis, M. Sc., F. de Ciencias, UNAM, 131 p. 1980.

Martin Del Pozzo, A.L., Monogenetic Vulcanism in Sierra Chichinautzin, Mexico, Bull. Volcanol., 45-1, 1982.

McElhinny, M. W., and R. T. Merril, Geomagnetic secular variation over the past 5 m.y. Rev. Geophys. Space Phys., 13, 687-708, 1975.

McFadden, P. L. and M. W. McElhinny, A physical model for paleosecular variation, Geophys. J. R. Astron. Soc. 78, 809-830, 1984.

McWiliams, M. O., Holcomb, R.T. and D. E. Champion, Geomagnetic secular variation from ^{14}C-dated lava flows on Hawaii and the question of the Pacific non-dipole low, Phil. Trans. R. Soc. Lond. A 306, 211-222, 1982.

Mooser, F., A. Nairn and J. F. W. Negendank, Paleomagnetic Investigations of the Tertiary and Quaternary igneous rocks: VIII A palaeomagnetic and petrologic study of volcanics of the Valley of Mexico, Geol. Rundsh., 63, 451-483, 1974.

Nixon, G. T., A. Demant, R. L. Armstrong, and J. E. Harakal, K-Ar and Geologic data bearing on the age and evolution of the Trans-Mexican Volcanic Belt, Geofísica Internacional, (in press), 1986.

Steele, W. K., Paleomagnetic constraints on the volcanic history of Iztaccihuatl, Geofísica Internacional, 24-1, 159-167, 1985.

Urrutia-Fucugauchi, J., Comments on Paleomagnetic constraints on the volcanic history of Iztaccihuatl, Geofísica Internacional, (in press), 1986.

U.S.-Japan Paleomagnetic Cooperation Program in Micronesia, Paleosecular variation of lavas from the Marianas in the Western Pacific Ocean, J. Geomag. Geoelectr., 27, 57-66, 1975.

(Received September 15, 1986;
Accepted October 13, 1986.)

19448007, 1986, 13, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL013i013p01442 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License